Chapter 26 Review

Campbell Biology, AP Edition, 12e

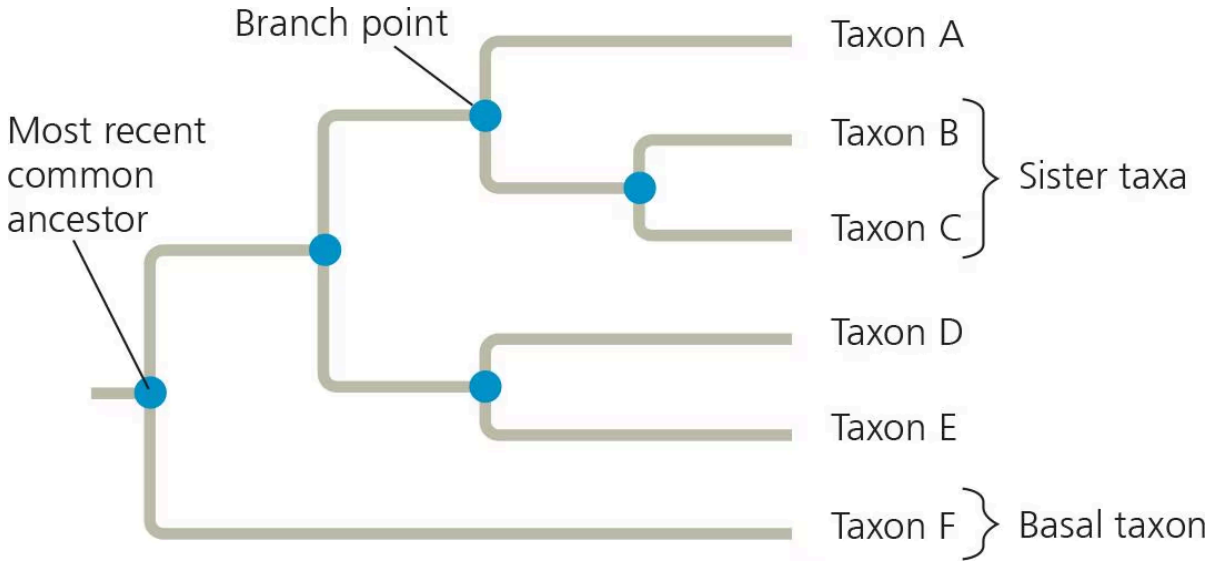
Chapter 26 Review

Summary of Key Concepts

Could the early tree of life be a ring? What evidence could be used to support a circular lineage for early life? **(BIG IDEAS 1 & 3)**

Concept 26.1: Phylogenies show evolutionary relationships

- Linnaeus's **binomial** classification system gives organisms two-part names: a **genus** plus a specific epithet.
- In the Linnaean system, species are grouped in increasingly broad **taxa**: Related genera are placed in the same **family**, families in **orders**, orders in **classes**, classes in **phyla**, phyla in ****kingdoms**, and (more recently) kingdoms in **domains**.
- Systematists depict evolutionary relationships as branching **phylogenetic trees**. Many systematists propose that classification be based entirely on evolutionary relationships.



The tree has branches that begin to the left. The first node is labeled Most recent common ancestor. It branches to Taxon F (basal taxon) and to another branch. That other branch goes to 2 more branch points. One branch point goes to taxons D and E. Another branches yet again. It goes to Taxon A and another branch. This final branch point goes to taxons B and C, which are sister taxa.

- Unless branch lengths are proportional to time or genetic change, a phylogenetic tree indicates only patterns of descent.
- Much information can be learned about a species from its evolutionary history; hence, phylogenies are useful in a wide range of applications.

Humans and chimpanzees are sister species. Explain what this statement means.

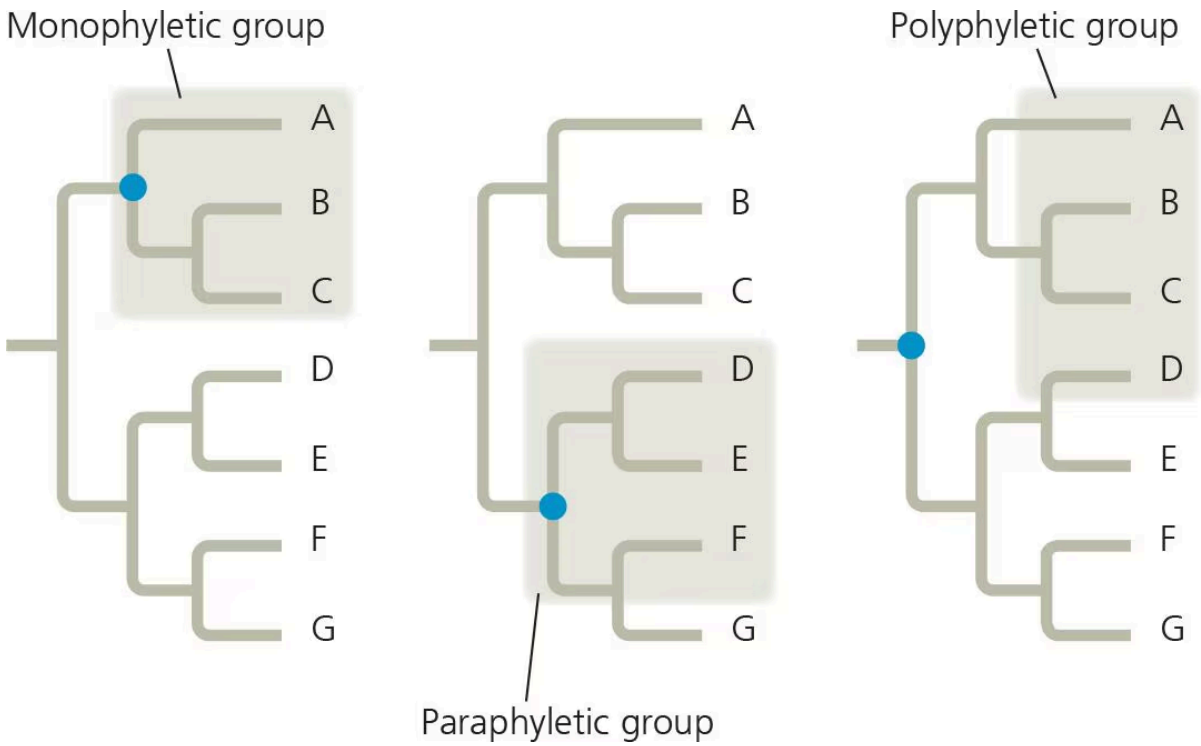
Concept 26.2: Phylogenies are inferred from morphological and molecular data

- Organisms with similar morphologies or DNA sequences are likely to be more closely related than organisms with very different structures and genetic sequences.
- To infer phylogeny, **homology** (similarity due to shared ancestry) must be distinguished from **analogy** (similarity due to convergent evolution).
- Computer programs are used to align comparable DNA sequences and to distinguish molecular homologies from coincidental matches between taxa that diverged long ago.

Why is it necessary to distinguish homology from analogy to infer phylogeny?

Concept 26.3: Shared characters are used to construct phylogenetic trees

- A **clade** is a monophyletic group that includes an ancestral species and all of its descendants.



A tree has 1 common branch. In one direction, it goes to a branch point that leads to taxon A and 2 another branch. The other branch goes to taxa B and C. Taxa A, B, and C are a monophyletic group. Returning to the common branch, it splits to a branch point. This point goes to 2 more branch points. One splits into taxa D and E, and the other splits to taxa F and G. Taxa D, E, and F are a paraphyletic group. In the third highlighted part, taxa A, B, C, and D are a polyphyletic group.

- Clades can be distinguished by their **shared derived characters**.

- Among phylogenies, the most parsimonious tree is the one that requires the fewest evolutionary changes. The most likely tree is the one based on the most likely pattern of changes.
- Well-supported phylogenetic hypotheses are consistent with a wide range of data.

Explain the logic of using shared derived characters to infer phylogeny.

Concept 26.4: An organism's evolutionary history is documented in its genome

- **Orthologous genes** are homologous genes found in different species as a result of speciation. **Paralogous genes** are homologous genes within a species that result from gene duplication; such genes can diverge and potentially take on new functions.
- Distantly related species often have many orthologous genes. The small variation in gene number in organisms of varying complexity suggests that genes are versatile and may have multiple functions.

When reconstructing phylogenies, is it more useful to compare orthologous or paralogous genes? Explain.

Concept 26.5: Molecular clocks help track evolutionary time

- Some regions of DNA change at a rate consistent enough to serve as a **molecular clock**, a method of estimating the date of past evolutionary events based on the amount of genetic change. Other DNA regions change in a less predictable way.
- Molecular clock analyses suggest that the most common strain of HIV jumped from primates to humans in the early 1900s.

Describe some assumptions and limitations of molecular clocks.

Concept 26.6: Our understanding of the tree of life continues to change based on new data

- Past classification systems have given way to the current view of the tree of life, which consists of three great domains: Bacteria, Archaea, and Eukarya.
- Phylogenies based in part on rRNA genes suggest that eukaryotes are most closely related to archaea, while data from some other genes suggest a closer relationship to bacteria.
- Genetic analyses indicate that extensive **horizontal gene transfer** has occurred throughout the evolutionary history of life.

Why was the five-kingdom system abandoned for a three-domain system?